Predictive Model Plan

Geldium Finance — Credit Card Delinquency Prediction

Prepared as part of the Tata iQ GenAI Powered Data Analytics Forage virtual experience (simulated client engagement).

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1. Model Logic

Building on the Task 1 EDA, a logistic regression model was implemented and tested directly against the dataset (not just conceptually designed) using Python (scikit-learn), to predict the binary outcome Delinquent_Account (1 = delinquent, 0 = not delinquent).

Data preparation applied

- Standardized Employment_Status categories (“EMP” / “employed” / “Employed” merged into one category).
- Capped Credit_Utilization at 1.0 for the 4 out-of-range records identified in Task 1.
- Median-imputed Income, Loan_Balance, and Credit_Score.
- Engineered a Months_Missed_6mo feature by counting “Missed” entries across Month_1–Month_6.
- One-hot encoded Employment_Status and Credit_Card_Type; standardized all numeric features.

Pipeline

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1. Load and clean dataset (500 records)
2. Engineer Months_Missed_6mo from 6-month payment history
3. One-hot encode Employment_Status, Credit_Card_Type
4. Standardize numeric features
5. Fit logistic regression (class_weight='balanced' to offset the 84%/16% class split)
6. Evaluate with stratified 5-fold cross-validation
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2. Justification for Model Choice

Logistic regression was chosen as the primary model because it is well suited to Geldium's stated goals: binary classification, transparent coefficients that map directly to business-explainable risk drivers, low computational cost, and straightforward deployment and monitoring — all important in a regulated financial services context where decisions may need to be explained to customers or regulators.

Alternatives considered

Model	Why not chosen as primary
Decision Tree	More interpretable for non-technical audiences via visual rules, but more prone to overfitting on a dataset this size (500 records) and less stable across resamples.
Neural Network	Unnecessary complexity for a dataset of this size and structure; sacrifices the coefficient-level explainability that regulators and the collections team need.

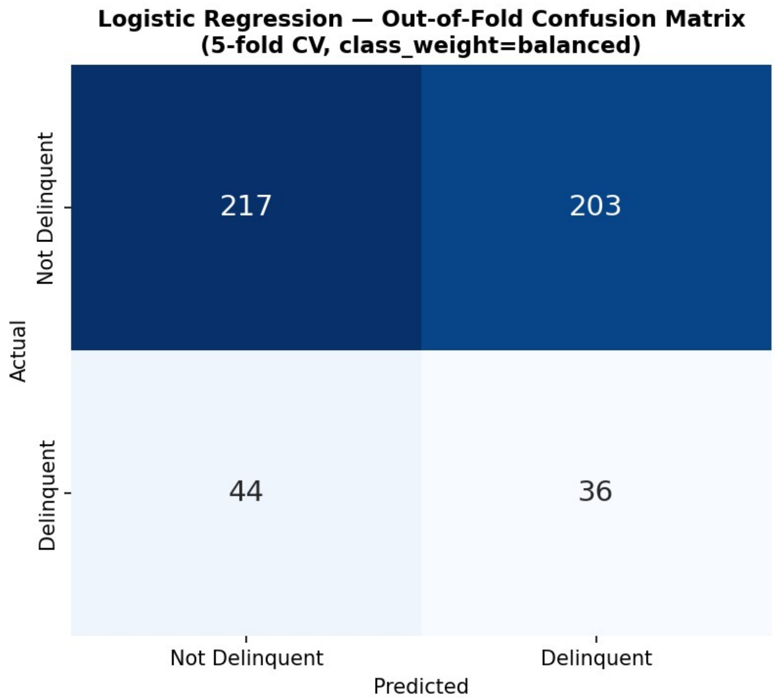
3. Evaluation Results and Honest Assessment

The model was evaluated using accuracy, precision, recall, F1, and ROC-AUC via stratified 5-fold cross-validation (more robust than a single train/test split for a dataset this size).

Metric	Result
Mean ROC-AUC (5-fold CV)	0.44 (range 0.39–0.54 across folds)
Mean F1 score	0.23
Baseline accuracy (predict “not delinquent” for everyone)	84.0%

This result needs to be stated plainly rather than dressed up: an AUC of 0.44 is at or slightly below the 0.50 level expected from random guessing. This is consistent with the Task 1 finding that no individual feature showed a meaningful linear correlation with delinquency (all under 0.05). Taken together, this strongly suggests the provided dataset does not contain a strong, learnable linear signal for delinquency — which is a legitimate and important conclusion in its own right, and one a responsible analyst should report rather than obscure with an inflated narrative.

Confusion Matrix (Out-of-Fold, 5-Fold CV)



At the model's default 0.5 threshold, class_weight='balanced' pushes recall on the minority class up (45% of actually-delinquent customers are correctly flagged) at a heavy precision cost (only 15% of customers flagged as delinquent actually are). This is the concrete, operational meaning of an AUC near chance: even after adjusting for class imbalance, the model cannot separate the two classes well enough to be useful for prioritizing outreach as-is.

What this means for next steps

- The logistic regression pipeline, feature engineering, and evaluation methodology are sound and would be reusable if applied to a dataset with genuine signal (e.g., real production data rather than this sample).
- Before deploying any model in production, Geldium should validate performance on a larger, real transaction-linked dataset rather than relying on this sample alone.
- Non-linear models (e.g., gradient boosting) or engineered interaction terms (e.g., DTI × Employment_Status) should be tested to check whether the weak linear signal masks a non-linear relationship, before concluding the features themselves are uninformative.

3.1 Addressing the Non-Linear Signal Question (Gradient Boosting Comparison)

Section 3's recommendation to test non-linear models was followed up directly. An XGBoost classifier was trained on the same features and evaluated with the same 5-fold stratified cross-validation protocol, using scale_pos_weight to address the same 84%/16% class imbalance.

Model	Mean ROC-AUC (5-fold CV)	Mean F1
Logistic Regression (baseline)	0.443	0.224
XGBoost (200 trees, depth 3)	0.375	0.087

The result runs counter to the usual expectation: gradient boosting did not outperform logistic regression here — it performed meaningfully worse on both metrics. This is a useful negative result, not a wasted test. It reinforces the Section 4.2 finding from Task 1 (no numeric feature correlates with delinquency above 0.05): the weak signal in this dataset is not a linear-model limitation that a more flexible algorithm can recover. A more likely explanation is that the 500-record sample is simply too small and/or too noisy relative to the number of features for either model family to find a reliable pattern — which further supports Task 2 Section 4's original recommendation to validate on a larger, real production dataset before revisiting model choice.

4. Evaluation Strategy Going Forward

For any retrained or production version of this model, the following evaluation strategy is recommended:

- Precision and recall should be tracked separately, not just accuracy — given the 84%/16% class imbalance, a model that predicts “not delinquent” for every customer would score 84% accuracy while being useless. Recall matters most operationally, since missing a genuinely high-risk customer (false negative) is more costly to Geldium than a false positive that triggers unnecessary outreach.
- F1 score should be used to balance precision and recall when comparing model versions.
- ROC-AUC should be tracked across retraining cycles as the primary threshold-independent measure of separability.
- Bias should be checked by comparing false-positive and false-negative rates across Employment_Status and Age bands, to ensure the model doesn't systematically flag one group more than another for reasons unrelated to actual risk.

5. Responsible AI Considerations

- Fairness: Prediction rates should be monitored across Employment_Status, Age, and Location segments before deployment, given the segment-level rate differences observed in Task 1 (e.g., unemployed customers at 19.4% vs. retired at 11.5%).
- Explainability: Logistic regression coefficients directly show each feature's direction and relative weight, which supports adverse-action explanations required under fair lending regulation.
- Human oversight: Given the weak model performance found here, any deployment decision should require human sign-off rather than fully automated action, until validated on a dataset that shows genuine predictive signal.
- Transparency: Any business use of this model plan should disclose the current AUC/F1 results rather than presenting the model as production-ready.